

LECTURE NOTES

Hard Problems in Cryptography

Mathematical Foundations for Post-Quantum Cryptography

Factorization · Discrete Logarithm · Lattices · Codes · Isogenies · Multivariate · Hash

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Classical hardness assumptions and post-quantum reductions

Hard Problems in Cryptography

Mathematical Foundations for Classical and Post-Quantum Cryptography

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Chapter 1

Preliminaries

1.1 Search vs. decision vs. distinguishing.

In post-quantum cryptography (PQC), many security assumptions are stated using one of three of hard problems:

1. **Search problem:** recover the hidden object.
2. **Decision problem:** answer a yes/no question about the instance.
3. **Distinguishing problem:** tell whether samples come from one distribution or another.

These three look similar, but they play different roles in theory and in cryptographic reductions.

Type	**Lattice**	**Code**
Search	SVP, CVP, BDD, SIS, Search-LWE	Syndrome Decoding, decoding, low-weight codeword search
Decision	GapSVP, GapCVP	existence of error of weight $\leq w$, gap decoding forms
Distinguishing	Decision-LWE, Ring-LWE, Module-LWE	structured code vs. random code, disguised secret code vs. random linear code

1.2 Preliminaries for Lattice-Based Hard Problems

This section collects the mathematical background needed to understand lattices, lattice basis reduction, and the hard problems that arise in post-quantum cryptography.

1.2.1 Notation

$\mathbb{Z}$ the ring of integers.

$\mathbb{R}$ the field of real numbers.

$\mathbb{Z}_q = \mathbb{Z}/q\mathbb{Z}$ integers modulo q.

$\mathbb{R}^n$ the n-dimensional real vector space.

$\mathbf{x}, \mathbf{y}, \mathbf{b}$ vectors.

A, B matrices.

$\langle \mathbf{x}, \mathbf{y} \rangle$ the standard inner product.

$\|\mathbf{x}\|$ the Euclidean norm of $\mathbf{x}$.

$\det(B)$ determinant of a square matrix B.

$\mathcal{L}(B)$ the lattice generated by the columns of B.

1.2.2 Vector spaces and Euclidean geometry

Vectors in $\mathbb{R}^n$

Definition 1.2.1. A vector in $\mathbb{R}^n$ is an ordered n-tuple

$$\mathbf{x} = (x_1, x_2, \ldots, x_n), \qquad x_i \in \mathbb{R}.$$

The set of all such vectors forms a vector space over $\mathbb{R}$.

Inner product

Definition 1.2.2. The standard inner product of two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ is defined by

$$\langle \mathbf{x}, \mathbf{y} \rangle = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n.$$

This quantity measures the geometric alignment of two vectors.

Remark 1.2.3. Two vectors are said to be *orthogonal* if

$$\langle \mathbf{x}, \mathbf{y} \rangle = 0.$$

Euclidean norm

Definition 1.2.4. The Euclidean norm of a vector $\mathbf{x} \in \mathbb{R}^n$ is

$$\|\mathbf{x}\| = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle} = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}.$$

Remark 1.2.5. This norm satisfies the standard properties:

1. $\|\mathbf{x}\| \geq 0$, with equality if and only if $\mathbf{x} = \mathbf{0}$,
2. $\|\alpha \mathbf{x}\| = |\alpha| \, \|\mathbf{x}\|$ for $\alpha \in \mathbb{R}$,
3. $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$ (triangle inequality).

Linear Independence

Definition 1.2.6. A finite set of vectors

$$\{\mathbf{b}_1, \ldots, \mathbf{b}_m\} \subseteq \mathbb{R}^n$$

is **linearly independent** if

$$a_1 \mathbf{b}_1 + \cdots + a_m \mathbf{b}_m = \mathbf{0}$$

implies

$$a_1 = \cdots = a_m = 0.$$

Remark 1.2.7 (span)**.** The real span of $\mathbf{b}_1, \ldots, \mathbf{b}_m$ is

$$V = \operatorname{span}_{\mathbb{R}}(\mathbf{b}_1, \ldots, \mathbf{b}_m) = \left\{ \sum_{i=1}^{m} a_i \mathbf{b}_i : a_i \in \mathbb{R} \right\}.$$

So V is just the set of all vectors you can build from $\mathbf{b}_1, \ldots, \mathbf{b}_m$ using real coefficients.

Remark 1.2.8. Let B be the matrix whose columns are $\mathbf{b}_1, \ldots, \mathbf{b}_m$:

$$B = \begin{bmatrix} | & & | \\ \mathbf{b}_1 & \cdots & \mathbf{b}_m \\ | & & | \end{bmatrix}.$$

Then

$$B\mathbf{a} = \begin{bmatrix} | & & | \\ \mathbf{b}_1 & \cdots & \mathbf{b}_m \\ | & & | \end{bmatrix} \begin{bmatrix} a_1 \\ \vdots \\ a_m \end{bmatrix} = a_1 \begin{bmatrix} | \\ \mathbf{b}_1 \\ | \end{bmatrix} + \cdots + a_m \begin{bmatrix} | \\ \mathbf{b}_m \\ | \end{bmatrix} = a_1 \mathbf{b}_1 + \cdots + a_m \mathbf{b}_m.,$$

So

$$V = \operatorname{span}_{\mathbb{R}}\{\mathbf{b}_1, \ldots, \mathbf{b}_m\} = \left\{ \sum_{i=1}^{m} a_i \mathbf{b}_i : a_i \in \mathbb{R} \right\} = \{B\mathbf{a} : \mathbf{a} \in \mathbb{R}^m\}.$$

The first writes the span as a sum. The second writes the same thing using matrix multiplication.

Lattice

Definition 1.2.9. A lattice $\Lambda \subset \mathbb{R}^n$ is called a lattice of rank m if

$$\Lambda = \left\{ \sum_{i=1}^{m} z_i \mathbf{b}_i : z_i \in \mathbb{Z} \right\},$$

where $\mathbf{b}_1, \ldots, \mathbf{b}_m \in \mathbb{R}^n$ are linearly independent.

Remark 1.2.10.

- If $m = n$, then Λ is called a full-rank lattice.
- If $m < n$, then Λ is a non-full-rank lattice, and it lies in the m-dimensional subspace

$$\operatorname{span}_{\mathbb{R}}\{\mathbf{b}_1, \ldots, \mathbf{b}_m\} \subsetneq \mathbb{R}^n.$$

Remark 1.2.11 (Full-rank Lattice)**.** A **full-rank lattice** $\Lambda \subset \mathbb{R}^n$ is a set of the form

$$\Lambda = \mathcal{L}(B) = \{B\mathbf{z} : \mathbf{z} \in \mathbb{Z}^n\},$$

where $B \in \mathbb{R}^{n \times n}$ is an invertible matrix. If $B = \begin{bmatrix} \mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_n \end{bmatrix}$, then equivalently

$$\mathcal{L}(B) = \{z_1\mathbf{b}_1 + \cdots + z_n\mathbf{b}_n : z_1, \ldots, z_n \in \mathbb{Z}\}.$$

Thus, a lattice is the set of all integer linear combinations of n linearly independent vectors in $\mathbb{R}^n$.

1.2.3 Orthogonality and Gram–Schmidt Orthogonalization

1.2.4 Fundamental Parallelepipeds and Determinants

Fundamental Parallelepiped

Let

$$B = \begin{bmatrix} | & | & \cdots & | \\ \mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_n \\ | & | & \cdots & | \end{bmatrix} \in \mathbb{R}^{n \times n}$$

be a basis matrix for a full-rank lattice

$$\Lambda = \mathcal{L}(B) = \{B\mathbf{z} : \mathbf{z} \in \mathbb{Z}^n\}.$$

Thus the columns $\mathbf{b}_1, \ldots, \mathbf{b}_n \in \mathbb{R}^n$ are linearly independent, and every lattice point is an integer linear combination of these basis vectors.

Fundamental Parallelepiped

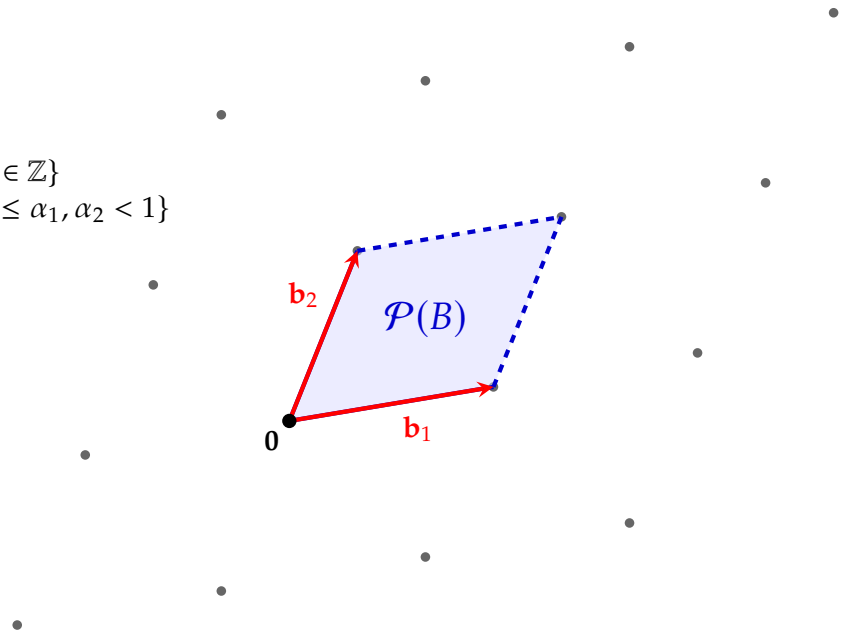
Definition 1.2.12. The **fundamental parallelepiped** associated with B is defined by

$$\mathcal{P}(B) = \left\{ \sum_{i=1}^{n} \alpha_i \mathbf{b}_i : 0 \le \alpha_i < 1 \right\}.$$

Equivalently, in matrix form,

$$\mathcal{P}(B) = \{B\boldsymbol{\alpha} : \boldsymbol{\alpha} \in [0,1)^n\}.$$

$\Lambda = \{z_1\mathbf{b}_1 + z_2\mathbf{b}_2 : z_1, z_2 \in \mathbb{Z}\}$
$\mathcal{P}(B) = \{\alpha_1\mathbf{b}_1 + \alpha_2\mathbf{b}_2 : 0 \le \alpha_1, \alpha_2 < 1\}$



The set $\mathcal{P}(B)$ is the higher-dimensional analogue of:

- an interval in dimension 1,
- a parallelogram in dimension 2,
- a parallelepiped in dimension 3.

Tiling Property

Proposition 1.2.13. *Let $B = [\mathbf{b}_1 \ \cdots \ \mathbf{b}_n] \in \mathbb{R}^{n\times n}$ be invertible, and let*

$$\Lambda = \mathcal{L}(B) = \{B\mathbf{z} : \mathbf{z} \in \mathbb{Z}^n\}$$

be the lattice generated by the columns of B. Define the fundamental parallelepiped

$$\mathcal{P}(B) = \left\{\sum_{i=1}^{n} \alpha_i \mathbf{b}_i : 0 \le \alpha_i < 1\right\} = \{B\boldsymbol{\alpha} : \boldsymbol{\alpha} \in [0,1)^n\}.$$

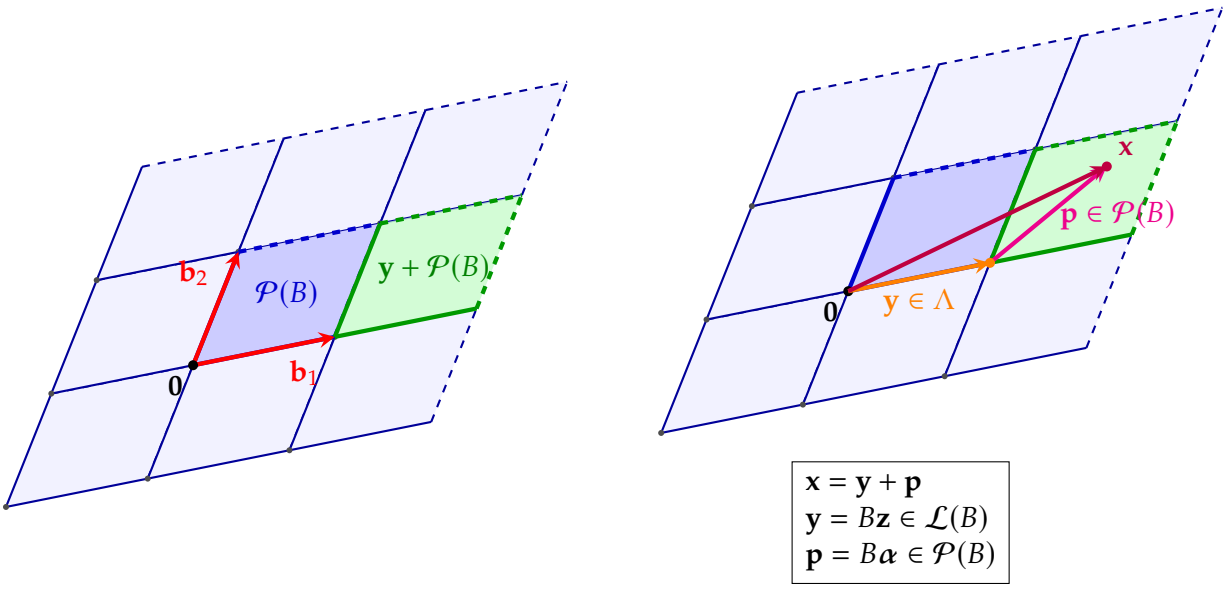
Then:

1. *$\mathbb{R}^n$ is covered by translates of $\mathcal{P}(B)$ by lattice vectors:*

$$\mathbb{R}^n = \bigcup_{\mathbf{y}\in\Lambda} (\mathbf{y} + \mathcal{P}(B)).$$

2. *If $\mathbf{y}_1, \mathbf{y}_2 \in \Lambda$ with $\mathbf{y}_1 \neq \mathbf{y}_2$, then*

$$(\mathbf{y}_1 + \mathcal{P}(B)) \cap (\mathbf{y}_2 + \mathcal{P}(B)) = \varnothing.$$



Proof. Since B is invertible, for every $\mathbf{x} \in \mathbb{R}^n$ there exists a unique vector $\mathbf{t} \in \mathbb{R}^n$ such that

$$\mathbf{x} = B\mathbf{t}.$$

Write

$$\mathbf{t} = \begin{pmatrix} t_1 \\ \vdots \\ t_n \end{pmatrix}.$$

For each i, let

$$z_i = \lfloor t_i \rfloor \in \mathbb{Z}, \qquad \alpha_i = t_i - z_i.$$

Then

$$0 \le \alpha_i < 1 \qquad \text{for each } i,$$

and $t_i = z_i + \alpha_i$. Hence

$$\mathbf{t} = \mathbf{z} + \boldsymbol{\alpha}, \quad \text{where} \quad \mathbf{z} = \begin{pmatrix} z_1 \\ \vdots \\ z_n \end{pmatrix} \in \mathbb{Z}^n, \qquad \boldsymbol{\alpha} = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} \in [0,1)^n.$$

Multiplying by B, we obtain

$$\mathbf{x} = B\mathbf{t} = B\mathbf{z} + B\boldsymbol{\alpha}.$$

Now $B\mathbf{z} \in \Lambda$, and $B\boldsymbol{\alpha} \in \mathcal{P}(B)$. Therefore

$$\mathbf{x} \in B\mathbf{z} + \mathcal{P}(B) \subseteq \bigcup_{\mathbf{y}\in\Lambda} (\mathbf{y} + \mathcal{P}(B)).$$

Since $\mathbf{x} \in \mathbb{R}^n$ was arbitrary, it follows that

$$\mathbb{R}^n = \bigcup_{\mathbf{y}\in\Lambda} (\mathbf{y} + \mathcal{P}(B)).$$

It remains to prove disjointness. Suppose

$$\mathbf{x} \in (\mathbf{y}_1 + \mathcal{P}(B)) \cap (\mathbf{y}_2 + \mathcal{P}(B))$$

for some $\mathbf{y}_1, \mathbf{y}_2 \in \Lambda$. Then there exist

$$\boldsymbol{\alpha}, \boldsymbol{\beta} \in [0,1)^n$$

such that

$$\mathbf{x} = \mathbf{y}_1 + B\boldsymbol{\alpha} = \mathbf{y}_2 + B\boldsymbol{\beta}.$$

Since $\mathbf{y}_1, \mathbf{y}_2 \in \Lambda$, there exist $\mathbf{z}_1, \mathbf{z}_2 \in \mathbb{Z}^n$ such that

$$\mathbf{y}_1 = B\mathbf{z}_1, \qquad \mathbf{y}_2 = B\mathbf{z}_2.$$

Therefore

$$B\mathbf{z}_1 + B\boldsymbol{\alpha} = B\mathbf{z}_2 + B\boldsymbol{\beta},$$

so

$$B(\mathbf{z}_1 - \mathbf{z}_2) = B(\boldsymbol{\beta} - \boldsymbol{\alpha}).$$

Because B is invertible,

$$\mathbf{z}_1 - \mathbf{z}_2 = \boldsymbol{\beta} - \boldsymbol{\alpha}.$$

The only integer in $(-1, 1)$ is 0, so

$$\boldsymbol{\beta} - \boldsymbol{\alpha} = \mathbf{0}.$$

Hence

$$\boldsymbol{\alpha} = \boldsymbol{\beta} \qquad \text{and} \qquad \mathbf{z}_1 = \mathbf{z}_2.$$

Therefore

$$\mathbf{y}_1 = B\mathbf{z}_1 = B\mathbf{z}_2 = \mathbf{y}_2.$$

This proves that if $\mathbf{y}_1 \neq \mathbf{y}_2$, then

$$(\mathbf{y}_1 + \mathcal{P}(B)) \cap (\mathbf{y}_2 + \mathcal{P}(B)) = \varnothing.$$

□

Uniqueness of Representation Modulo the Lattice

Theorem 1.2.14. *Let $B \in \mathbb{R}^{n\times n}$ be invertible, and let*

$$\Lambda = \mathcal{L}(B) = \{B\mathbf{z} : \mathbf{z} \in \mathbb{Z}^n\}$$

be the lattice generated by the columns of B. Let

$$\mathcal{P}(B) = \{B\boldsymbol{\alpha} : \boldsymbol{\alpha} \in [0,1)^n\}$$

be the associated fundamental parallelepiped. Then every vector $\mathbf{x} \in \mathbb{R}^n$ can be written uniquely in the form

$$\mathbf{x} = \mathbf{y} + \mathbf{p}, \qquad \mathbf{y} \in \mathcal{L}(B),\ \mathbf{p} \in \mathcal{P}(B).$$

Equivalently, for every $\mathbf{x} \in \mathbb{R}^n$, there exist unique $\mathbf{z} \in \mathbb{Z}^n$, $\boldsymbol{\alpha} \in [0,1)^n$ such that

$$\mathbf{x} = B\mathbf{z} + B\boldsymbol{\alpha} = B(z + \boldsymbol{\alpha}).$$

Proof. Let $\mathbf{x} \in \mathbb{R}^n$ be arbitrary. Since B is invertible, there exists a unique vector

$$\mathbf{t} = \begin{pmatrix} t_1 \\ \vdots \\ t_n \end{pmatrix} \in \mathbb{R}^n$$

such that $\mathbf{x} = B\mathbf{t}$.

(Existence) For each coordinate t_i, define

$$z_i = \lfloor t_i \rfloor \in \mathbb{Z}, \qquad \alpha_i = t_i - z_i.$$

Then

$$0 \le \alpha_i < 1 \qquad \text{for each } i,$$

and $t_i = z_i + \alpha_i$. Hence

$$\mathbf{t} = \mathbf{z} + \boldsymbol{\alpha}, \quad \text{where} \quad \mathbf{z} = \begin{pmatrix} z_1 \\ \vdots \\ z_n \end{pmatrix} \in \mathbb{Z}^n, \qquad \boldsymbol{\alpha} = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} \in [0,1)^n.$$

Multiplying by B, we obtain

$$\mathbf{x} = B\mathbf{t} = B(\mathbf{z} + \boldsymbol{\alpha}) = B\mathbf{z} + B\boldsymbol{\alpha}.$$

Now $B\mathbf{z} \in \Lambda$ by definition of the lattice, and $B\boldsymbol{\alpha} \in \mathcal{P}(B)$ by definition of the fundamental parallelepiped. Therefore

$$\mathbf{x} = \mathbf{y} + \mathbf{p}$$

with

$$\mathbf{y} = B\mathbf{z} \in \Lambda, \qquad \mathbf{p} = B\boldsymbol{\alpha} \in \mathcal{P}(B).$$

(Uniqueness) Suppose that

$$\mathbf{x} = \mathbf{y}_1 + \mathbf{p}_1 = \mathbf{y}_2 + \mathbf{p}_2,$$

where

$$\mathbf{y}_1, \mathbf{y}_2 \in \Lambda, \qquad \mathbf{p}_1, \mathbf{p}_2 \in \mathcal{P}(B).$$

Since $\mathbf{y}_1, \mathbf{y}_2 \in \Lambda$, there exist $\mathbf{z}_1, \mathbf{z}_2 \in \mathbb{Z}^n$ such that

$$\mathbf{y}_1 = B\mathbf{z}_1, \qquad \mathbf{y}_2 = B\mathbf{z}_2.$$

Since $\mathbf{p}_1, \mathbf{p}_2 \in \mathcal{P}(B)$, there exist $\boldsymbol{\alpha}, \boldsymbol{\beta} \in [0,1)^n$ such that

$$\mathbf{p}_1 = B\boldsymbol{\alpha}, \qquad \mathbf{p}_2 = B\boldsymbol{\beta}.$$

Thus

$$B\mathbf{z}_1 + B\boldsymbol{\alpha} = B\mathbf{z}_2 + B\boldsymbol{\beta}.$$

Rearranging gives

$$B(\mathbf{z}_1 - \mathbf{z}_2) = B(\boldsymbol{\beta} - \boldsymbol{\alpha}).$$

Since B is invertible,

$$\mathbf{z}_1 - \mathbf{z}_2 = \boldsymbol{\beta} - \boldsymbol{\alpha}.$$

The only integer in $(-1, 1)$ is 0, so

$$\boldsymbol{\beta} - \boldsymbol{\alpha} = \mathbf{0}.$$

Hence

$$\boldsymbol{\alpha} = \boldsymbol{\beta} \qquad \text{and} \qquad \mathbf{z}_1 = \mathbf{z}_2.$$

Therefore

$$\mathbf{y}_1 = B\mathbf{z}_1 = B\mathbf{z}_2 = \mathbf{y}_2$$

and

$$\mathbf{p}_1 = B\boldsymbol{\alpha} = B\boldsymbol{\beta} = \mathbf{p}_2.$$

So the representation is unique. □

Determinant of a Full-rank Lattice

Let $\Lambda = \mathcal{L}(B)$ be a full-rank lattice in $\mathbb{R}^n$, where $B \in \mathbb{R}^{n \times n}$ is invertible. The *determinant* of the lattice is defined by

$$\det(\mathcal{L}(B)) = |\det(B)|.$$

Example 1.2.15. If

$$B = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix},$$

then

$$\det(B) = 2, \qquad \det(\mathcal{L}(B)) = 2.$$

Hence the area of the fundamental parallelogram is 2.

Discriminant / volume expression

Invariance under Basis Change

Relation to Short Vectors.

1.2.5 Successive Minima and Shortest Vectors

Chapter 2

Lattice-Based Hard Problems

A (full-rank) lattice $\Lambda \subset \mathbb{R}^n$ is

$$\Lambda = \{B\mathbf{z} : \mathbf{z} \in \mathbb{Z}^n\}$$

for some basis matrix $B \in \mathbb{R}^{n \times n}$. Classic algorithmic problems are

- **SVP (Shortest Vector Problem):** Find $\mathbf{0} \neq \mathbf{v} \in \Lambda$ minimizing $\|\mathbf{v}\|_2$.
- **GapSVP (Decision/SVP approximation):** Given (Λ, d) decide whether $\lambda_1(\Lambda) \leq d$ or $\lambda_1(\Lambda) > \gamma d$.
- **SIVP (Shortest Independent Vectors):** Find n linearly independent vectors of length $\leq \gamma \cdot \lambda_n(\Lambda)$.

The importance for cryptography: average-case problems (LWE/SIS) reduce from worst-case lattice problems (GapSVP/SIVP) under suitable parameters.

2.1 Learning With Errors (LWE)

TBA

2.2 Short Integer Solution (SIS)

TBA

2.3 NTRU Search Problem

TBA

Appendix A

Exercises for Lattice

A.1 Euclidean spaces and lattices

1. Let
$$\mathbf{x} = (1, -2, 3), \qquad \mathbf{y} = (4, 0, -1).$$
Compute $\langle \mathbf{x}, \mathbf{y} \rangle$, $\|\mathbf{x}\|$, $\|\mathbf{y}\|$, and $\|\mathbf{x} + \mathbf{y}\|$.

Sol.
$$\begin{aligned}
\langle \mathbf{x}, \mathbf{y} \rangle &= (1)(4) + (-2)(0) + (3)(-1) = 4 + 0 - 3 = 1. \\
\|\mathbf{x}\| &= \sqrt{1^2 + (-2)^2 + 3^2} = \sqrt{1 + 4 + 9} = \sqrt{14}. \\
\|\mathbf{y}\| &= \sqrt{4^2 + 0^2 + (-1)^2} = \sqrt{16 + 0 + 1} = \sqrt{17}. \\
\mathbf{x} + \mathbf{y} &= (1 + 4, -2 + 0, 3 + (-1)) = (5, -2, 2), \\
\|\mathbf{x} + \mathbf{y}\| &= \sqrt{5^2 + (-2)^2 + 2^2} = \sqrt{25 + 4 + 4} = \sqrt{33}.
\end{aligned}$$
Therefore,
$$\boxed{\langle \mathbf{x}, \mathbf{y} \rangle = 1, \qquad \|\mathbf{x}\| = \sqrt{14}, \qquad \|\mathbf{y}\| = \sqrt{17}, \qquad \|\mathbf{x} + \mathbf{y}\| = \sqrt{33}.}$$
□

2. Determine whether the following sets are lattices in $\mathbb{R}^2$:
   (a) $\mathbb{Z}^2$
   (b) $\{(a, \sqrt{2}a) : a \in \mathbb{Z}\}$
   (c) $\{a(1, 0) + b(1/2, 1) : a, b \in \mathbb{Z}\}$
   (d) $\mathbb{Q}^2$

Sol. (a) Yes, this is a lattice. Indeed,
$$\mathbb{Z}^2 = \{a(1, 0) + b(0, 1) : a, b \in \mathbb{Z}\} = \mathcal{L}(\mathbf{e}_1, \mathbf{e}_2).$$
(b) Yes, this is a lattice of rank 1 in $\mathbb{R}^2$. We may write it as
$$\{a(1, \sqrt{2}) : a \in \mathbb{Z}\} = \mathcal{L}(1, \sqrt{2}).$$

(c) Yes, this is a lattice in $\mathbb{R}^2$.

$$\{a(1,0) + b(1/2,1) : a, b \in \mathbb{Z}\} = \mathcal{L}(\mathbf{b}_1, \mathbf{b}_2) \quad \text{where} \quad \mathbf{b}_1 = (1,0), \qquad \mathbf{b}_2 = \left(\frac{1}{2}, 1\right).$$

(d) No, this is not a lattice. There are rational points arbitrarily close to $(0,0)$, for example

$$\left(\frac{1}{n}, 0\right) \in \mathbb{Q}^2 \qquad \text{for all } n \in \mathbb{N},$$

and these converge to $(0,0)$. Hence $\mathbb{Q}^2$ is not a lattice.

Therefore:

$$\boxed{\mathbb{Z}^2,\ \{(a, \sqrt{2}a) : a \in \mathbb{Z}\},\ \{a(1,0) + b(1/2,1) : a, b \in \mathbb{Z}\} \text{ are lattices, but } \mathbb{Q}^2 \text{ is not.}}$$

□

3. Let

$$\mathbf{b}_1 = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \qquad \mathbf{b}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Describe explicitly the real span of $\mathbf{b}_1, \mathbf{b}_2$, and the lattice generated by them.

Sol. The real span is

$$\operatorname{span}_{\mathbb{R}}\{\mathbf{b}_1, \mathbf{b}_2\} = \left\{ a \begin{pmatrix} 2 \\ 0 \end{pmatrix} + b \begin{pmatrix} 1 \\ 1 \end{pmatrix} : a, b \in \mathbb{R} \right\}.$$

Computing,

$$a \begin{pmatrix} 2 \\ 0 \end{pmatrix} + b \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2a + b \\ b \end{pmatrix}.$$

Given any $(x, y) \in \mathbb{R}^2$, choose $b = y$ and $a = (x - y)/2$. Hence every vector in $\mathbb{R}^2$ has this form, so

$$\operatorname{span}_{\mathbb{R}}\{\mathbf{b}_1, \mathbf{b}_2\} = \mathbb{R}^2.$$

The lattice generated by $\mathbf{b}_1, \mathbf{b}_2$ is

$$\Lambda = \left\{ m \begin{pmatrix} 2 \\ 0 \end{pmatrix} + n \begin{pmatrix} 1 \\ 1 \end{pmatrix} : m, n \in \mathbb{Z} \right\} = \left\{ \begin{pmatrix} 2m + n \\ n \end{pmatrix} : m, n \in \mathbb{Z} \right\}.$$

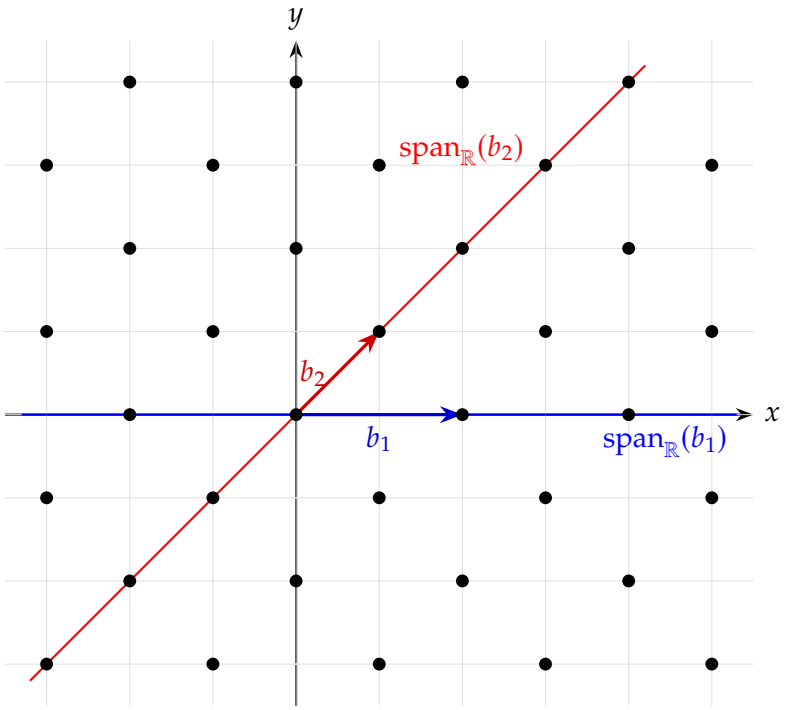
Equivalently,

$$\Lambda = \{(x, y) \in \mathbb{Z}^2 : x - y \in 2\mathbb{Z}\} = \{(x, y) \in \mathbb{Z}^2 : x \equiv y \pmod 2\}.$$

So the lattice consists of all integer points whose coordinates have the same parity. Therefore,

$$\boxed{\operatorname{span}_{\mathbb{R}}\{\mathbf{b}_1, \mathbf{b}_2\} = \mathbb{R}^2, \qquad \Lambda = \left\{ \begin{pmatrix} 2m + n \\ n \end{pmatrix} : m, n \in \mathbb{Z} \right\}.}$$

□



4. Show that if $\mathbf{b}_1, \ldots, \mathbf{b}_m \in \mathbb{R}^n$ are linearly independent, then

$$\Lambda = \left\{ \sum_{i=1}^{m} z_i \mathbf{b}_i : z_i \in \mathbb{Z} \right\}$$

is a discrete subgroup of $\mathbb{R}^n$.

Sol. We first show that Λ is a subgroup of $(\mathbb{R}^n, +)$. If

$$\mathbf{x} = \sum_{i=1}^{m} z_i \mathbf{b}_i, \qquad \mathbf{y} = \sum_{i=1}^{m} w_i \mathbf{b}_i$$

are in Λ, with $z_i, w_i \in \mathbb{Z}$, then

$$\mathbf{x} - \mathbf{y} = \sum_{i=1}^{m} (z_i - w_i) \mathbf{b}_i.$$

Since $z_i - w_i \in \mathbb{Z}$, we have $\mathbf{x} - \mathbf{y} \in \Lambda$.

Let

$$V = \mathrm{span}_{\mathbb{R}}\{\mathbf{b}_1, \ldots, \mathbf{b}_m\}.$$

Since the vectors are linearly independent, the map

$$T : \mathbb{R}^m \to V, \qquad T(z_1, \ldots, z_m) = \sum_{i=1}^{m} z_i \mathbf{b}_i$$

is a linear isomorphism. In particular, $T(\mathbb{Z}^m) = \Lambda$. Because $\mathbb{Z}^m$ is discrete in $\mathbb{R}^m$, and T is a homeomorphism from $\mathbb{R}^m$ onto V, it follows that Λ is discrete in V, hence also discrete in $\mathbb{R}^n$. Therefore Λ is a discrete subgroup of $\mathbb{R}^n$. □

5. Give two distinct bases of the same lattice in $\mathbb{R}^2$.

Sol. Consider the standard lattice

$$\mathbb{Z}^2.$$

One basis is

$$\mathbf{b}_1 = (1,0), \qquad \mathbf{b}_2 = (0,1).$$

Another basis is

$$\mathbf{c}_1 = (1,1), \qquad \mathbf{c}_2 = (1,0).$$

Indeed,

$$\mathbf{c}_1 = (1,1), \qquad \mathbf{c}_2 = (1,0)$$

are linearly independent, and every integer vector $(x, y) \in \mathbb{Z}^2$ can be written as

$$(x, y) = y(1,1) + (x - y)(1,0).$$

Since $x, y \in \mathbb{Z}$, this is an integer linear combination of $\mathbf{c}_1, \mathbf{c}_2$. Hence

$$\mathbb{Z}^2 = \mathcal{L}(\mathbf{b}_1, \mathbf{b}_2) = \mathcal{L}(\mathbf{c}_1, \mathbf{c}_2).$$

Thus two distinct bases of the same lattice are

$$\boxed{\{(1,0),(0,1)\} \qquad \text{and} \qquad \{(1,1),(1,0)\}.}$$

□

6. Let $B = [\mathbf{b}_1\ \mathbf{b}_2]$ with

$$\mathbf{b}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \qquad \mathbf{b}_2 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}.$$

Write $\mathcal{L}(B)$ explicitly.

Sol. The lattice generated by $\mathbf{b}_1, \mathbf{b}_2$ is

$$\mathcal{L}(B) = \{z_1\mathbf{b}_1 + z_2\mathbf{b}_2 : z_1, z_2 \in \mathbb{Z}\}.$$

Substituting the basis vectors,

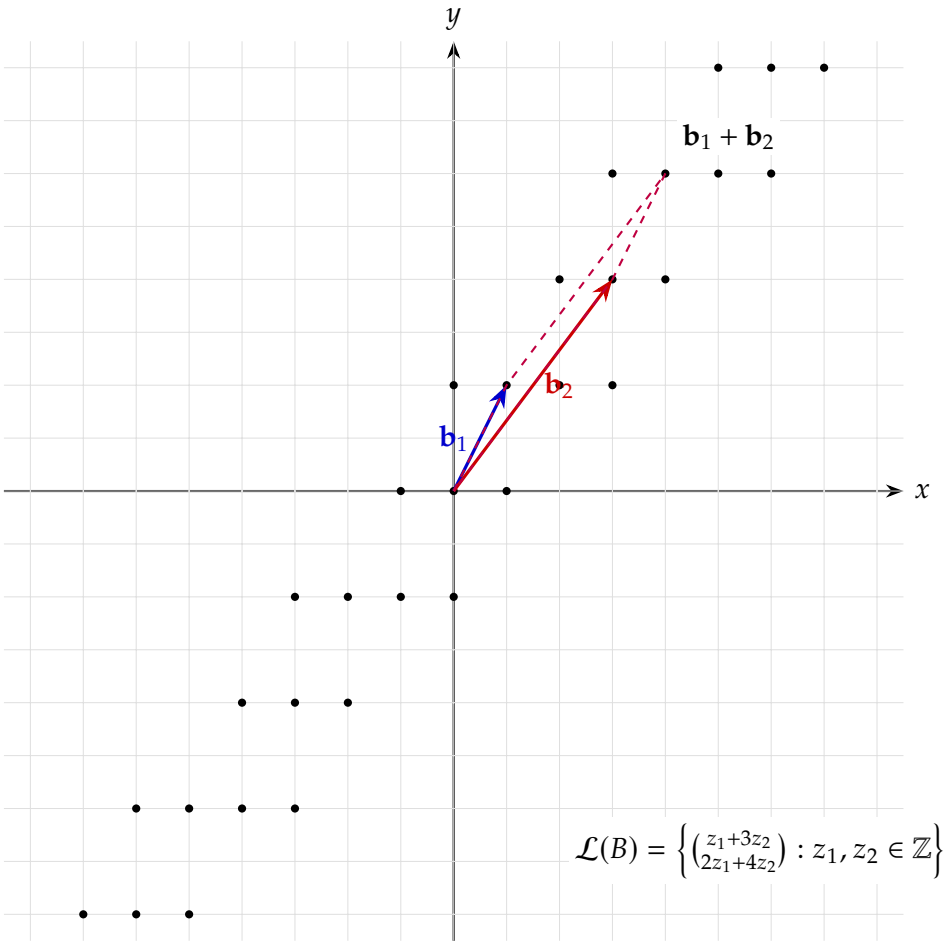
$$\mathcal{L}(B) = \left\{ z_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + z_2 \begin{pmatrix} 3 \\ 4 \end{pmatrix} : z_1, z_2 \in \mathbb{Z} \right\}.$$

Thus

$$\mathcal{L}(B) = \left\{ \begin{pmatrix} z_1 + 3z_2 \\ 2z_1 + 4z_2 \end{pmatrix} : z_1, z_2 \in \mathbb{Z} \right\}.$$

So explicitly,

$$\boxed{\mathcal{L}(B) = \left\{(x, y) \in \mathbb{R}^2 : x = z_1 + 3z_2,\ y = 2z_1 + 4z_2,\ z_1, z_2 \in \mathbb{Z}\right\}.}$$



□

7. Show that a lattice may have infinitely many bases.

Sol. Consider the lattice $\mathbb{Z}^2$. For each integer $k \in \mathbb{Z}$, define

$$\mathbf{b}_1^{(k)} = (1, 0), \qquad \mathbf{b}_2^{(k)} = (k, 1).$$

Then

$$B_k = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$$

has $\det(B_k) = 1$, so its columns form a basis of $\mathbb{Z}^2$. Indeed, every vector in $\mathcal{L}(B_k)$ has the form

$$a(1, 0) + b(k, 1) = (a + bk, b), \qquad a, b \in \mathbb{Z}.$$

Given any $(x, y) \in \mathbb{Z}^2$, choose $b = y$ and $a = x - ky$. Then

$$(x, y) = a(1, 0) + b(k, 1),$$

so $\mathcal{L}(B_k) = \mathbb{Z}^2$. As k varies over $\mathbb{Z}$, these bases are distinct. Therefore the lattice $\mathbb{Z}^2$ has infinitely many bases. Hence a lattice may have infinitely many bases. □

8. Let $\Lambda \subseteq \mathbb{R}^n$ be a lattice. Prove that Λ is countable.

Sol. Suppose

$$\Lambda = \left\{ \sum_{i=1}^{m} z_i \mathbf{b}_i : z_i \in \mathbb{Z} \right\}$$

for some linearly independent vectors $\mathbf{b}_1, \ldots, \mathbf{b}_m \in \mathbb{R}^n$. Define the map

$$f : \mathbb{Z}^m \to \Lambda, \qquad f(z_1, \ldots, z_m) = \sum_{i=1}^{m} z_i \mathbf{b}_i.$$

This map is surjective by definition of Λ. Since $\mathbb{Z}^m$ is countable, and the image of a countable set under a map is countable, it follows that Λ is countable. Therefore every lattice is countable. □

9. Show that every lattice contains the zero vector and is closed under addition and negation.

Sol. Let

$$\Lambda = \left\{ \sum_{i=1}^{m} z_i \mathbf{b}_i : z_i \in \mathbb{Z} \right\}.$$

First, Λ contains the zero vector because taking all coefficients equal to 0 gives

$$\mathbf{0} = \sum_{i=1}^{m} 0 \cdot \mathbf{b}_i \in \Lambda.$$

Now let

$$\mathbf{x} = \sum_{i=1}^{m} z_i \mathbf{b}_i, \qquad \mathbf{y} = \sum_{i=1}^{m} w_i \mathbf{b}_i$$

be vectors in Λ. Then

$$\mathbf{x} + \mathbf{y} = \sum_{i=1}^{m} (z_i + w_i)\mathbf{b}_i.$$

Since $z_i + w_i \in \mathbb{Z}$, we have $\mathbf{x} + \mathbf{y} \in \Lambda$. Thus Λ is closed under addition. Also,

$$-\mathbf{x} = \sum_{i=1}^{m} (-z_i)\mathbf{b}_i.$$

Since $-z_i \in \mathbb{Z}$, it follows that $-\mathbf{x} \in \Lambda$. Thus Λ is closed under negation. Therefore every lattice contains $\mathbf{0}$ and is closed under addition and negation. □

10. Let

$$\Lambda = \{(2a + b, b) : a, b \in \mathbb{Z}\}.$$

Find a basis matrix B such that $\Lambda = \mathcal{L}(B)$.

Sol. We rewrite the general element of Λ as

$$\Lambda = \{a(2, 0) + b(1, 1) : a, b \in \mathbb{Z}\}.$$

So a basis matrix is obtained by taking these vectors as columns:

$$B = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}.$$

Then

$$B \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 2a + b \\ b \end{pmatrix}.$$

Therefore

$$\boxed{B = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}} \qquad \text{and} \qquad \Lambda = \mathcal{L}(B).$$

□

A.2 Gram–Schmidt Orthogonalization

A.3 Fundamental Parallelepipeds and Determinants

Recall the standard decomposition

$$\mathbb{R}^n = \bigcup_{\mathbf{z}\in\mathbb{Z}^n} \big(\mathbf{z} + [0,1)^n\big), \qquad \big(\mathbf{z}_1 + [0,1)^n\big) \cap \big(\mathbf{z}_2 + [0,1)^n\big) = \varnothing \quad (\mathbf{z}_1 \neq \mathbf{z}_2).$$

Use the invertible linear map $B : \mathbb{R}^n \to \mathbb{R}^n$ to transport this statement to the lattice

$$\Lambda = \mathcal{L}(B) = \{B\mathbf{z} : \mathbf{z} \in \mathbb{Z}^n\}$$

and the parallelepiped

$$\mathcal{P}(B) = \{B\boldsymbol{\alpha} : \boldsymbol{\alpha} \in [0,1)^n\}.$$

21-24 Let

$$B = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}, \qquad \Lambda = \mathcal{L}(B), \qquad \mathcal{P}(B) = \{B\boldsymbol{\alpha} : \boldsymbol{\alpha} \in [0,1)^2\}.$$

21. Compute B^{-1}.

22. For each of the following vectors $\mathbf{x}$, compute

$$\mathbf{t} = B^{-1}\mathbf{x}, \qquad \mathbf{z} = \lfloor \mathbf{t} \rfloor \in \mathbb{Z}^2, \qquad \boldsymbol{\alpha} = \mathbf{t} - \lfloor \mathbf{t} \rfloor \in [0,1)^2,$$

and hence find the unique decomposition

$$\mathbf{x} = B\mathbf{z} + B\boldsymbol{\alpha}.$$

Use:

$$\begin{pmatrix} 7.4 \\ 5.2 \end{pmatrix}, \qquad \begin{pmatrix} -1.3 \\ 4.7 \end{pmatrix}, \qquad \begin{pmatrix} 3 \\ -2.4 \end{pmatrix}, \qquad \begin{pmatrix} -5.8 \\ -0.2 \end{pmatrix}.$$

23. For each vector above, identify explicitly the lattice vector

$$\mathbf{y} = B\mathbf{z} \in \Lambda$$

and the point

$$\mathbf{p} = B\boldsymbol{\alpha} \in \mathcal{P}(B)$$

such that $\mathbf{x} = \mathbf{y} + \mathbf{p}$.

24. Check in each case that $\mathbf{p} \in \mathcal{P}(B)$ by writing it in the form

$$\mathbf{p} = \alpha_1\mathbf{b}_1 + \alpha_2\mathbf{b}_2 \qquad \text{with } 0 \le \alpha_1, \alpha_2 < 1.$$

25-28 Let

$$B = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}, \qquad \Lambda = \mathcal{L}(B), \qquad \mathcal{P}(B) = \{B\boldsymbol{\alpha} : \boldsymbol{\alpha} \in [0,1)^2\}.$$

25. Compute the four vertices of $\mathcal{P}(B)$.

26. For each point $\mathbf{x}$ below, determine the unique lattice vector $\mathbf{y} \in \Lambda$ such that

$$\mathbf{x} \in \mathbf{y} + \mathcal{P}(B).$$

Also compute the corresponding $\mathbf{p} \in \mathcal{P}(B)$ with

$$\mathbf{x} = \mathbf{y} + \mathbf{p}.$$

Use:

$$\begin{pmatrix} 4.2 \\ 5.1 \end{pmatrix}, \qquad \begin{pmatrix} -0.7 \\ 1.6 \end{pmatrix}, \qquad \begin{pmatrix} 3.9 \\ -2.4 \end{pmatrix}, \qquad \begin{pmatrix} -3.3 \\ -4.8 \end{pmatrix}.$$

27. For each point, verify your answer by computing $B^{-1}\mathbf{x}$ and separating its coordinates into integer parts and fractional parts.

28. Pick one of the four points and draw the translate $\mathbf{y} + \mathcal{P}(B)$ that contains it.

29-33 Let

$$B = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 2 \end{pmatrix}, \qquad \Lambda = \mathcal{L}(B).$$

For $\mathbf{x} \in \mathbb{R}^3$, define its remainder modulo Λ by

$$r_B(\mathbf{x}) = B\big(\{B^{-1}\mathbf{x}\}\big),$$

where fractional parts are taken coordinatewise.

29. Compute $r_B(\mathbf{x})$ for each of the following vectors:

$$\mathbf{x}_1 = \begin{pmatrix} 5.2 \\ 3.7 \\ -1.4 \end{pmatrix}, \qquad \mathbf{x}_2 = \begin{pmatrix} -0.6 \\ 4.1 \\ 2.3 \end{pmatrix}, \qquad \mathbf{x}_3 = \begin{pmatrix} 7.2 \\ 5.7 \\ 0.6 \end{pmatrix}.$$

30. For each $\mathbf{x}_i$, also compute the corresponding lattice vector

$$\mathbf{y}_i = \mathbf{x}_i - r_B(\mathbf{x}_i) \in \Lambda.$$

31. Decide which pairs among $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ lie in the same coset modulo Λ. Justify your answer by comparing their remainders.

32. For every pair that lies in the same coset, compute explicitly the lattice vector

$$\mathbf{x}_i - \mathbf{x}_j \in \Lambda.$$

33. Explain how these calculations illustrate the uniqueness of the representation

$$\mathbf{x} = \mathbf{y} + \mathbf{p}, \qquad \mathbf{y} \in \Lambda,\ \mathbf{p} \in \mathcal{P}(B).$$

A.4 Successive Minima and Short Vector

A.5 Dual Lattices

A.6 Reduced Bases and the LLL algorithm

A.7 q-ary Lattices and SIS

A.8 Conceptual and Proof Exercises